

DBMS Lab 1

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Aim: To perform and understand fundamental DBMS operations, including database creation, table management, data manipulation, modification, and retrieval using MySQL.

SHOW

databases;

```
MySQL localhost:3306 ssl lakshay_match SQL > show databases;
+-----+
| Database |
+-----+
| information_schema |
| lakshay_match      |
| mysql              |
| performance_schema |
| shre51_match       |
| sys                 |
+-----+
```

CREATE DATABASE stud1_test;

```
MySQL localhost:3306 ssl sys SQL > CREATE DATABASE stud1_test;
Query OK, 1 row affected (0.0566 sec)
```

SHOW DATABASES;

```
MySQL localhost:3306 ssl sys SQL > SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| lakshay_match      |
| mysql              |
| performance_schema |
| shre51_match       |
| stud1_test         |
| sys                 |
+-----+
```

CREATE DATABASE stud1test;

```
MySQL localhost:3306 ssl sys SQL > CREATE DATABASE stud1test;
Query OK, 1 row affected (0.0037 sec)
```

SHOW DATABASES;

```
MySQL localhost:3306 ssl sys SQL > SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| lakshay_match |
| mysql |
| performance_schema |
| shre51_match |
| stud1_test |
| stud1test |
| sys |
+-----+
```

DROP DATABASE stud1test;

```
MySQL localhost:3306 ssl sys SQL > DROP DATABASE stud1test;
Query OK, 0 rows affected (0.0142 sec)
```

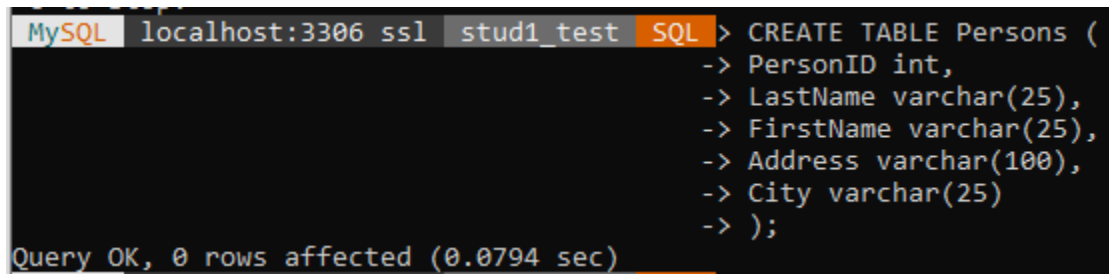
SHOW databases;

```
MySQL localhost:3306 ssl sys SQL > SHOW databases;
+-----+
| Database |
+-----+
| information_schema |
| lakshay_match |
| mysql |
| performance_schema |
| shre51_match |
| stud1_test |
| sys |
+-----+
```

USE stud1_test;

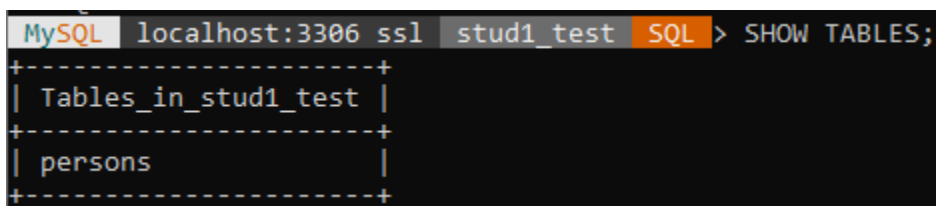
```
MySQL localhost:3306 ssl sys SQL > USE stud1_test;
Default schema set to `stud1_test`.
Fetching global names, object names from `stud1_test` for auto-completion... Press
^C to stop.
```

```
CREATE TABLE Persons (  
  
    PersonID int,  
  
    LastName varchar(25),  
  
    FirstName varchar(25),  
  
    Address varchar(100),  
  
    City varchar(25)  
  
);
```



```
MySQL localhost:3306 ssl stud1_test SQL > CREATE TABLE Persons (  
    -> PersonID int,  
    -> LastName varchar(25),  
    -> FirstName varchar(25),  
    -> Address varchar(100),  
    -> City varchar(25)  
    -> );  
Query OK, 0 rows affected (0.0794 sec)
```

```
SHOW TABLES;
```



```
MySQL localhost:3306 ssl stud1_test SQL > SHOW TABLES;  
+-----+  
| Tables_in_stud1_test |  
+-----+  
| persons               |  
+-----+
```

```
DESCRIBE Persons;
```

```
MySQL localhost:3306 ssl stud1_test SQL > DESCRIBE Persons;
```

Field	Type	Null	Key	Default	Extra
PersonID	int	YES		NULL	
LastName	varchar(25)	YES		NULL	
FirstName	varchar(25)	YES		NULL	
Address	varchar(100)	YES		NULL	
City	varchar(25)	YES		NULL	

DROP TABLE Persons;

```
MySQL localhost:3306 ssl stud1_test SQL > DROP TABLE Persons;
Query OK, 0 rows affected (0.0201 sec)
```

CREATE TABLE Customers (CustomerID int,LastName varchar(25),FirstName
varchar(25),Address varchar(100),City varchar(25), primary key
(CustomerID));

```
MySQL localhost:3306 ssl stud1_test SQL > CREATE TABLE Customers (CustomerID int,LastName varchar(25),FirstName
-> varchar(25),Address varchar(100),City varchar(25), primary key
-> (CustomerID));
Query OK, 0 rows affected (0.0185 sec)
```

DESCRIBE Customers;

```
MySQL localhost:3306 ssl stud1_test SQL > DESCRIBE Customers;
```

Field	Type	Null	Key	Default	Extra
CustomerID	int	NO	PRI	NULL	
LastName	varchar(25)	YES		NULL	
FirstName	varchar(25)	YES		NULL	
Address	varchar(100)	YES		NULL	
City	varchar(25)	YES		NULL	

INSERT INTO Customers (CustomerID,LastName,FirstName,Address,City)
VALUES (96365,'Shah','Hiren','A67,Building10,abc lane','Mumbai');

```
MySQL localhost:3306 ssl stud1_test SQL > INSERT INTO Customers (CustomerID,LastName,FirstName,Address,City)
-> VALUES (96365,'Shah','Hiren','A67,Building10,abc lane','Mumbai');
Query OK, 1 row affected (0.0131 sec)
```

SELECT * FROM Customers;

```
MySQL localhost:3306 ssl stud1_test SQL > SELECT * FROM Customers;
+-----+-----+-----+-----+-----+
| CustomerID | LastName | FirstName | Address | City |
+-----+-----+-----+-----+-----+
| 96365 | Shah | Hiren | A67,Building10,abc lane | Mumbai |
+-----+-----+-----+-----+-----+
```

INSERT INTO Customers VALUES

(96300,'Singh','Prakesh','B7,House10,abc lane','Mumbai'),

(89790,'Joshi','Ramesh','413,Sadashiv Nagar','Pune');

```
MySQL localhost:3306 ssl stud1_test SQL > INSERT INTO Customers VALUES
-> (96300,'Singh','Prakesh','B7,House10,abc lane','Mumbai'),
-> (89790,'Joshi','Ramesh','413,Sadashiv Nagar','Pune');
Query OK, 2 rows affected (0.0027 sec)
Records: 2 Duplicates: 0 Warnings: 0
```

```
MySQL localhost:3306 ssl stud1_test SQL > INSERT INTO Customers
-> (CustomerID, LastName, FirstName, Address, City)
-> VALUES
-> (63215, 'Sharma', 'Aarav', '12 MG Road', 'Mumbai'),
-> (74582, 'Patil', 'Sneha', '45 FC Road', 'Pune'),
-> (51839, 'Mehta', 'Riya', '78 C G Road', 'Ahmedabad'),
-> (92641, 'Verma', 'Aditya', '23 Hazratganj', 'Lucknow'),
-> (38176, 'Iyer', 'Ananya', '56 Anna Salai', 'Chennai'),
-> (85429, 'Reddy', 'Vikram', '91 Banjara Hills', 'Hyderabad'),
-> (46713, 'Gupta', 'Karan', '34 Connaught Place', 'New Delhi'),
-> (71958, 'Joshi', 'Neha', '27 MG Road', 'Bengaluru'),
-> (58324, 'Deshmukh', 'Omkar', '19 Civil Lines', 'Nagpur'),
-> (64197, 'Kulkarni', 'Pooja', '63 Law College Road', 'Nashik');
Query OK, 10 rows affected (0.0080 sec)
Records: 10 Duplicates: 0 Warnings: 0
```

----Now add approx. 10 customers-----

SELECT * FROM Customers;

```
MySQL localhost:3306 ssl stud1_test SQL > SELECT * FROM Customers;
```

CustomerID	LastName	FirstName	Address	City	Email	Remark
38176	Iyer	Ananya	56 Anna Salai	Chennai	NULL	NULL
46713	Gupta	Karan	34 Connaught Place	New Delhi	NULL	NULL
51839	Mehta	Riya	78 C G Road	Ahmedabad	NULL	NULL
58324	Deshmukh	Omkar	19 Civil Lines	Nagpur	NULL	NULL
63215	Sharma	Aarav	12 MG Road	Mumbai	NULL	NULL
64197	Kulkarni	Pooja	63 Law College Road	Nashik	NULL	NULL
71958	Joshi	Neha	27 MG Road	Bengaluru	NULL	NULL
74582	Patil	Sneha	45 FC Road	Pune	NULL	NULL
85429	Reddy	Vikram	91 Banjara Hills	Hyderabad	NULL	NULL
89790	Joshi	Ramesh	413,Sadashiv Nagar	Pune	ramesh.joshi@abc.com	Regular Buyer
92641	Verma	Aditya	23 Hazratganj	Lucknow	NULL	NULL
96300	Singh	Prakesh	B7,House10,abc lane	Mumbai	NULL	NULL
96365	Shah	Hiren	A67,Building10,abc lane	Mumbai	NULL	NULL

13 rows in set (0.0004 sec)

ALTER TABLE Customers ADD Email varchar(50);

```
MySQL localhost:3306 ssl stud1_test SQL > ALTER TABLE Customers ADD Email varchar(50);
```

ERROR: 1060 (42521): Duplicate column name 'Email'

```
MySQL localhost:3306 ssl stud1_test SQL > DESCRIBE Customers;
```

Field	Type	Null	Key	Default	Extra
CustomerID	int	NO	PRI	NULL	
LastName	varchar(25)	YES		NULL	
FirstName	varchar(25)	YES		NULL	
Address	varchar(100)	YES		NULL	
City	varchar(25)	YES		NULL	
Email	varchar(50)	YES		NULL	
Remark	varchar(255)	YES		NULL	

DESCRIBE Customers;

```
MySQL localhost:3306 ssl stud1_test SQL > DESCRIBE Customers;
```

Field	Type	Null	Key	Default	Extra
CustomerID	int	NO	PRI	NULL	
LastName	varchar(25)	YES		NULL	
FirstName	varchar(25)	YES		NULL	
Address	varchar(100)	YES		NULL	
City	varchar(25)	YES		NULL	
Email	varchar(50)	YES		NULL	
Remark	varchar(255)	YES		NULL	

ALTER TABLE Customers ADD Remark varchar(255);

```
MySQL localhost:3306 ssl stud1_test SQL > ALTER TABLE Customers ADD Remark varchar(255);
```

DESCRIBE Customers;

```
MySQL localhost:3306 ssl stud1_test SQL > DESCRIBE Customers;
```

Field	Type	Null	Key	Default	Extra
CustomerID	int	NO	PRI	NULL	
LastName	varchar(25)	YES		NULL	
FirstName	varchar(25)	YES		NULL	
Address	varchar(100)	YES		NULL	
City	varchar(25)	YES		NULL	
Email	varchar(50)	YES		NULL	
Remark	varchar(255)	YES		NULL	

SELECT * FROM Customers;

```
MySQL localhost:3306 ssl stud1_test SQL > SELECT * FROM Customers;
```

CustomerID	LastName	FirstName	Address	City	Email	Remark
38176	Iyer	Ananya	56 Anna Salai	Chennai	NULL	NULL
46713	Gupta	Karan	34 Connaught Place	New Delhi	NULL	NULL
51839	Mehta	Riya	78 C G Road	Ahmedabad	NULL	NULL
58324	Deshmukh	Omkar	19 Civil Lines	Nagpur	NULL	NULL
63215	Sharma	Aarav	12 MG Road	Mumbai	NULL	NULL
64197	Kulkarni	Pooja	63 Law College Road	Nashik	NULL	NULL
71958	Joshi	Neha	27 MG Road	Bengaluru	NULL	NULL
74582	Patil	Sneha	45 FC Road	Pune	NULL	NULL
85429	Reddy	Vikram	91 Banjara Hills	Hyderabad	NULL	NULL
89790	Joshi	Ramesh	413,Sadashiv Nagar	Pune	ramesh.joshi@abc.com	Regular Buyer
92641	Verma	Aditya	23 Hazratganj	Lucknow	NULL	NULL
96300	Singh	Prakesh	B7,House10,abc lane	Mumbai	NULL	NULL
96365	Shah	Hiren	A67,Building10,abc lane	Mumbai	NULL	NULL

UPDATE Customers SET Email='ramesh.joshi@abc.com' , Remark='Regular Buyer' WHERE

CustomerID=89790;

```
MySQL localhost:3306 ssl stud1_test SQL > UPDATE Customers SET Email='ramesh.joshi@abc.com' , Remark='Regular Buyer' WHERE
-> CustomerID=89790;
Query OK, 0 rows affected (0.0004 sec)
Rows matched: 1 Changed: 0 Warnings: 0
```

SELECT CustomerID,Address FROM Customers;

```
MySQL localhost:3306 ssl stud1_test SQL > SELECT CustomerID,Address FROM Customers;
```

CustomerID	Address
38176	56 Anna Salai
46713	34 Connaught Place
51839	78 C G Road
58324	19 Civil Lines
63215	12 MG Road
64197	63 Law College Road
71958	27 MG Road
74582	45 FC Road
85429	91 Banjara Hills
89790	413,Sadashiv Nagar
92641	23 Hazratganj
96300	B7,House10,abc lane
96365	A67,Building10,abc lane

----Try yourself on different set of attributes-----

SELECT CustomerID,Address FROM Customers WHERE CustomerID=96300;

```
MySQL localhost:3306 ssl stud1_test SQL > SELECT CustomerID,Address FROM Customers WHERE CustomerID=96300;
```

CustomerID	Address
96300	B7,House10,abc lane

SELECT * FROM Customers WHERE CustomerID=96300;

```
MySQL localhost:3306 ssl stud1_test SQL > SELECT * FROM Customers WHERE CustomerID=96300;
```

CustomerID	LastName	FirstName	Address	City	Email	Remark
96300	Singh	Prakesh	B7,House10,abc lane	Mumbai	NULL	NULL

1 row in set (0.0002 sec)

What is a database? Why do we need a database management system such as MySQL?

A database is an organized collection of data that can be stored, accessed, and managed easily. For example, a customer database may store customer names, IDs, phone numbers, and addresses.

A Database Management System (DBMS) such as MySQL is software used to create, store, retrieve, update, and manage data in databases. We need a DBMS because it makes handling large amounts of data easier, faster, more secure, and more organized.

What is the difference between CREATE DATABASE, SHOW DATABASES, and DROP DATABASE?

Command ->Purpose

CREATE DATABASE -> Creates a new database.

SHOW DATABASES -> Displays the databases available in MySQL.

DROP DATABASE -> Deletes an entire database and its contents.

What is a table in a database? Explain the purpose of rows and columns in a table.

A table is a structured collection of related data in a database. It consists of rows and columns.

- Columns represent the different attributes or fields of the data.
- Rows represent individual records.

What is a primary key? Why was CustomerID selected as the primary key in the Customers table?

A primary key is a column, or combination of columns, that uniquely identifies each row in a table. A primary key cannot contain duplicate values and normally cannot be NULL.

CustomerID is a suitable primary key for the Customers table because each customer can be assigned a unique ID. Unlike names or addresses, customer IDs can reliably distinguish one customer from another.

What is the purpose of the VARCHAR and INT data types? Give one suitable example of each.

VARCHAR is used to store variable-length text/string values.

Example:

Name VARCHAR(50)

This can store a customer's name, such as "Rahul Sharma".

INT is used to store whole numbers (integers).

Example:

Age INT

This can store a value such as 25.

What is the difference between INSERT, SELECT, UPDATE, and ALTER TABLE?

Command	Purpose
<u>INSERT</u>	Adds a new record/rows to a table.
<u>SELECT</u>	Retrieves or displays data from the table.
<u>UPDATE</u>	Changes the existing data in a table.
<u>ALTER TABLE</u>	Changes the structure of an existing table, such as adding or removing columns.

Why is the **WHERE** clause important in an **UPDATE** statement? What could happen if we write an **UPDATE** statement without a **WHERE** condition?

The **WHERE** clause specifies which rows should be updated. It is therefore extremely important when modifying existing data.

For example:

```
UPDATE Customers
```

```
SET Age = 26
```

```
WHERE CustomerID = 101;
```

Only the customer whose **CustomerID** is **101** will be updated.

If we write:

```
UPDATE Customers SET Age = 26;
```

without a **WHERE** condition, MySQL will update every row in the **Customers** table. This could accidentally change all customers' ages to **26**.

Therefore, when using **UPDATE**, always check that the **WHERE** condition identifies exactly the records you intend to change.